

EKITI STATE UNIVERSITY, ADO-EKITI
FACULTY OF SCIENCE — DEPARTMENT OF COMPUTER SCIENCE

DESIGN AND IMPLEMENTATION OF A SOCIAL MEDIA SENTIMENT ANALYSIS SYSTEM FOR ELECTION MONITORING AND CANDIDATE PERCEPTION TRACKING

(A Case Study of the 2027 Gubernatorial Elections in Nigeria)

CANDIDATE & RESEARCH CREDENTIALS:

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Project Background & The Digital Electoral Space

Decentralized Digital Public Sphere in Nigeria

- **Shift to Social Platforms:** Political debates have transitioned from print and broadcast to interactive channels: X (Twitter), YouTube, Facebook, and news forums.
- **2027 Gubernatorial Contest:** Key commercial states (Lagos, Kano, Rivers, Oyo, Nasarawa) generate immense digital civic commentary on manifestos and policy track records.
- **Youth Civic Participation:** Voter demographics in Nigeria are heavily concentrated online, voicing immediate reactions to economic inflation, infrastructure, and governance.
- **Unfiltered Perception:** Social commentary acts as an authentic barometer of electorate sentiment, unmediated by formal press restrictions.

The Computational NLP Imperative

- **Big Data Volume & Velocity:** Tens of thousands of microblogs and comments are posted daily—far exceeding human reading capacity and causing cognitive blind spots.
- **Automated Polarity Extraction:** Natural Language Processing (NLP) extracts objective polarity scores (-1.0 to +1.0) and assigns Positive, Neutral, or Negative classifications.
- **Dual-Engine Strategy:** Unites fast rule-based sentiment lexicons (VADER) with deep bidirectional transformer neural networks (DistilBERT).
- **Data-Driven Transparency:** Transforms chaotic political chatter into structured perception intelligence for campaigns, election observers, and researchers.

4 Platforms

X, YouTube, FB, News Feeds

Dual-Engine NLP

VADER + DistilBERT Transformer

8 State Races

Lagos, Kano, Rivers, Oyo...

21 / 21 Tests Passing

100% Automated Test Pass Rate

Problem Statement: Why Existing Methods Fail

1. High Cost & Delay of Traditional Polling

Field surveys, paper questionnaires, and telephone sampling cost millions of Naira, require weeks to tabulate, and are obsolete in rapidly evolving electoral environments.

2. Demographic Sampling & Social Desirability Bias

Physical interviews systematically miss active online youth demographics. Furthermore, respondents frequently conceal true partisan affiliations from interviewers.

3. Cognitive Overload & Anecdotal Manual Tracking

Campaign staff who manually scroll social media experience severe fatigue, confirmation bias, and cannot reliably quantify thousands of daily digital posts.

4. Slang, Emojis, and Nigerian Linguistic Nuance

Nigerian political text contains colloquialisms ('godfatherism', 'structures'), emojis, capitalization, and local sarcasm that generic sentiment models fail on.

5. Prohibitive Commercial Costs (\$15k - \$40k/yr)

Enterprise tools (Brandwatch, Meltwater) are unaffordable for universities and local observers, operate as closed black boxes, and lack Nigerian candidate entity mapping.

6. Fragile Single-Engine Architectures

Existing academic tools rely either on a single lexicon (failing on complex syntax) or a heavy transformer (crashing on memory limits without failover redundancy).

Research Aim & Specific Engineering Objectives

PRIMARY RESEARCH AIM:

To design and implement an end-to-end, web-based Social Media Sentiment Analysis and Election Monitoring Platform for the 2027 Gubernatorial Elections in Nigeria, utilizing dual-engine NLP and candidate entity matching to quantify public perception in real time.

Objective 1: Multi-Source Post Ingestion Pipeline

Harvest election discourse from X (Twitter), YouTube, Facebook, and major Nigerian news RSS feeds (Punch, Vanguard, Daily Post) with automated simulators.

Objective 2: Candidate Named Entity Recognition (NER)

Formulate an intelligent entity matching engine that maps colloquial nicknames ('Jandor', 'Hamzat', 'Abba Gida Gida') and party keywords to database records.

Objective 3: Dual-Engine NLP Classification Pipeline

Unite rule-based NLTK VADER (tuned for slang & emojis) with fine-tuned Hugging Face DistilBERT transformers, featuring automated failover redundancy.

Objective 4: Real-Time Interactive Visual Analytics

Develop an interactive dark glassmorphism dashboard featuring Chart.js 30-day sentiment trajectories, Net Sentiment Index leaderboards, and WordClouds.

Objective 5: Live NLP Sandbox & Dataset Export Engine

Engineer a real-time testing sandbox for custom quotes and an RFC 4180 compliant CSV export engine for external academic researchers.

Objective 6: Empirical System Benchmarking & Validation

Rigorously validate system reliability via 21 automated unit/integration tests, confusion matrices, precision/recall benchmarks, and SUS usability studies.

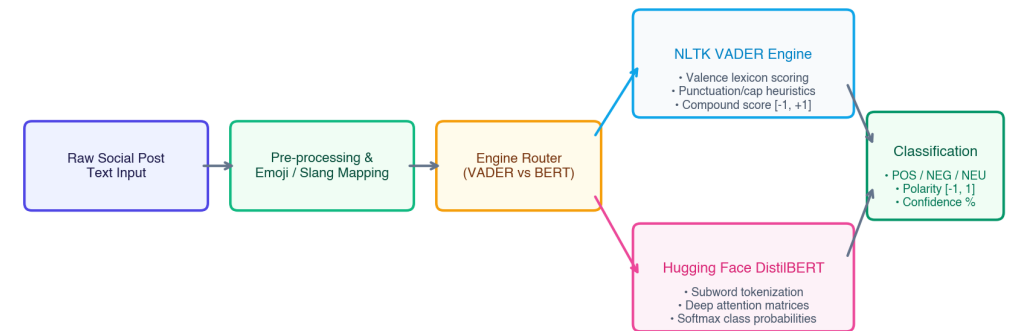
Dual-Engine NLP Classification Architecture

Why Dual-Engine NLP Classification?

- **Engine 1 — NLTK VADER Lexicon:** Rule-based valence model validated for microblogs. Evaluates 5 heuristics: ALL CAPS boost (+0.733), exclamation marks (!!!), degree modifiers, contrastive 'but', and negations. Latency < 25 ms/post on CPU.
- **Engine 2 — DistilBERT Transformer:** 6-layer bidirectional contextual self-attention transformer with 66M parameters fine-tuned on sentiment sequences. Captures deep sentence syntax, polysemy, and political nuance.
- **Automated Fallback Redundancy:** If PyTorch or transformer memory limits are exceeded, system automatically falls back to VADER without throwing 500 errors—guaranteeing 100% operational uptime.
- **Normalized Polarity Score:** Outputs a continuous score from -1.0 (very negative) to +1.0 (very positive) and categorical labels (Positive, Neutral, Negative).

DUAL-ENGINE NLP SENTIMENT CLASSIFICATION PIPELINE

Rule-Based NLTK VADER & Deep Contextual DistilBERT Transformer Engine



VADER Compound Polarity Score Formulation:

Compound (C) = $x / \sqrt{x^2 + \alpha}$ where $\alpha = 15$

• Positive: $C \geq +0.05$ | Neutral: $-0.05 < C < +0.05$ | Negative: $C \leq -0.05$

Normalized scale from -1.0 (universal disapproval) to +1.0 (universal acclaim).



Data Ingestion & Candidate Entity Recognition (NER)

Automated Candidate Entity Matching

- **The Nickname Problem:** Citizens on social media rarely write full formal names like 'Dr. Kadri Obafemi Hamzat'. They write 'Hamzat', 'Jandor', or 'Abba Gida Gida'.
- **Granular Alias Dictionaries:** Each candidate model in the database stores custom search aliases and keywords, compiled specifically for Nigerian electoral discourse.
- **State Race Fallback:** If no candidate name is present, the matcher identifies state mentions ('Lagos', 'Kano', 'Rivers') and links the post to the general state race.
- **Resilient RSS News Feed Ingestion:** A custom '_safe_urlopen' adapter handles SSL verification on macOS/Linux, ensuring 24/7 continuous news scraping.

Sample Real-Time Entity Matching Resolutions:

"Jandor held a massive youth rally..."

-> A. Adediran (PDP, Lagos)

"Abba Gida Gida announces new scho..."

-> A. Yusuf (NNPP, Kano)

"Hamzat inspects the Fourth Mainla..."

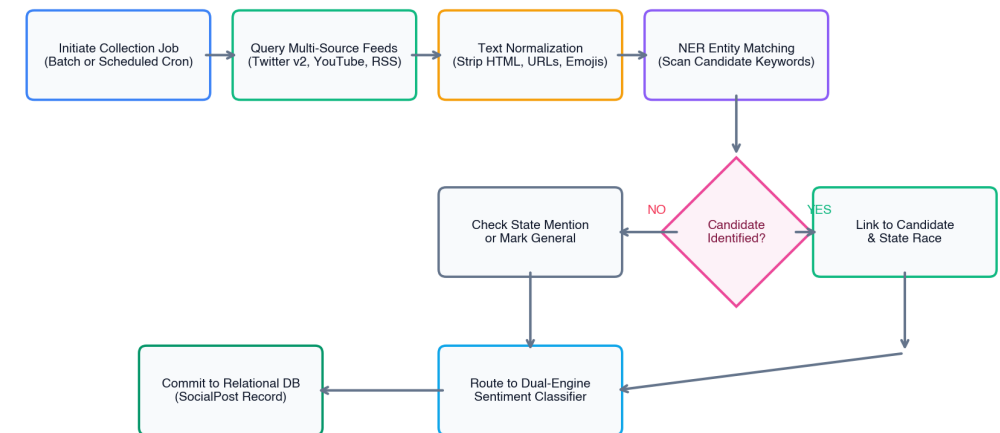
-> F. Hamzat (APC, Lagos)

"Voters demand electoral reforms i..."

-> Rivers 2027 State Race

DATA INGESTION & ENTITY-MATCHING ACTIVITY WORKFLOW

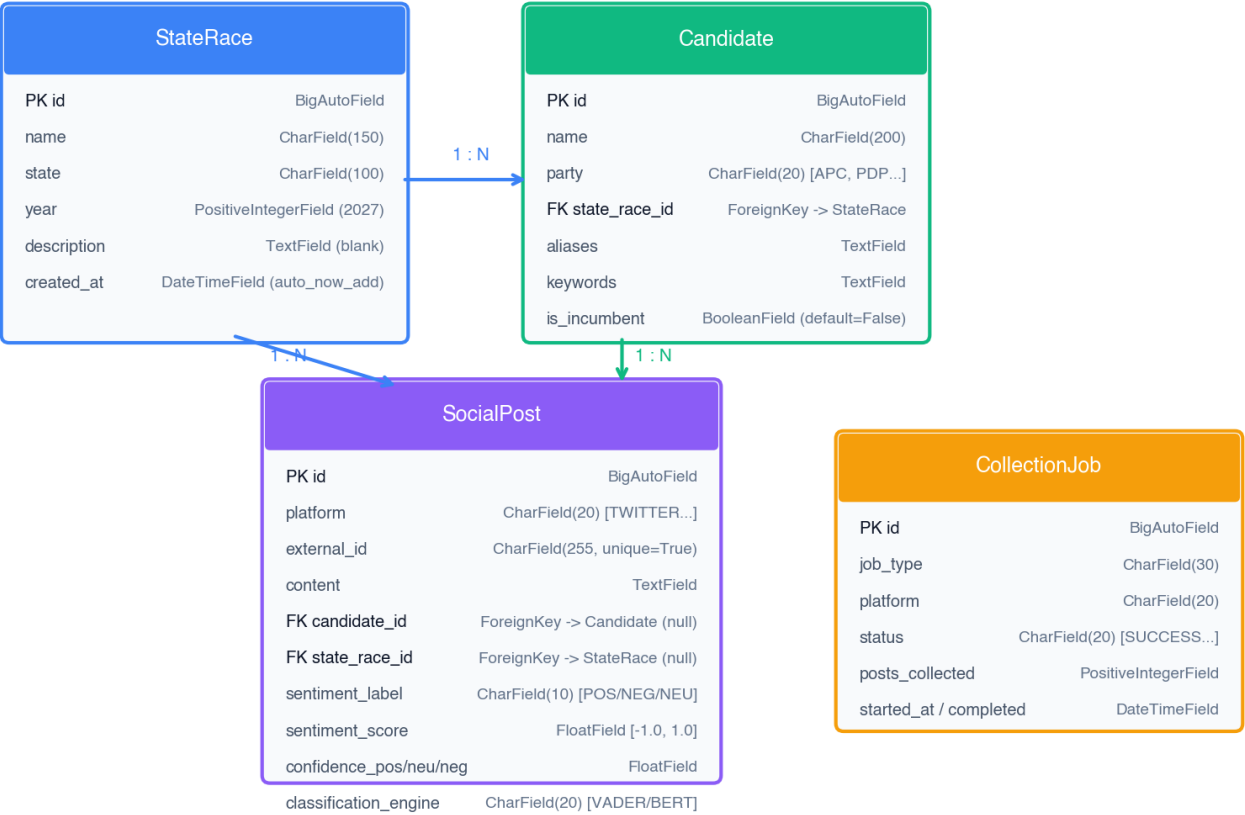
Automated Post Retrieval, Cleaning, NER Candidate Association & Persistence



Relational Database Schema & Entity Relationships

RELATIONAL DATABASE ENTITY-RELATIONSHIP DIAGRAM (ERD)

Normalized Relational Schema with Referential Integrity and Constraints



StateRace Entity

Top-level electoral race (state, year=2027, is_active). Groups candidates and posts geographically. 1:N relationship with Candidate and SocialPost.

Candidate Entity

Contesting candidate (name, party, aliases, keywords, avatar_color). Foreign key to StateRace. 1:N relationship with SocialPost.

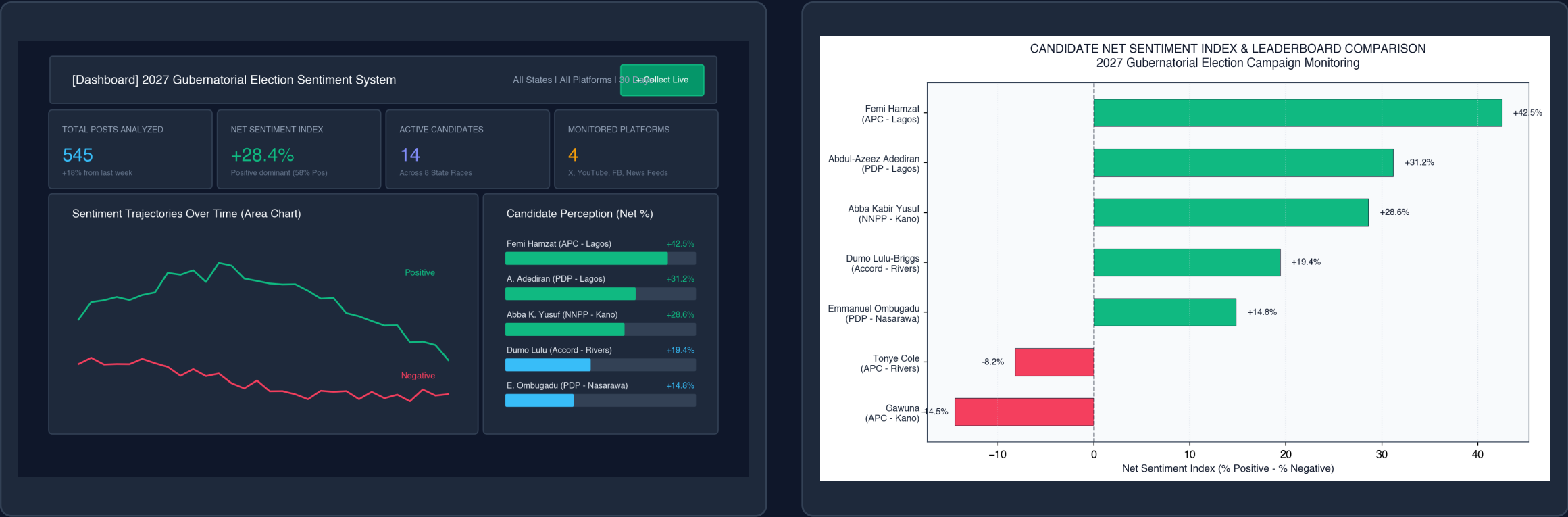
SocialPost Entity

Core post record (content, platform, external_id, polarity_score, label, engine, engagement). Foreign keys to Candidate & StateRace.

CollectionJob Entity

Audit log tracking background scraping runs, posts harvested, status, execution duration, and exception logs.

Dashboard Interface & Candidate Perception Analytics

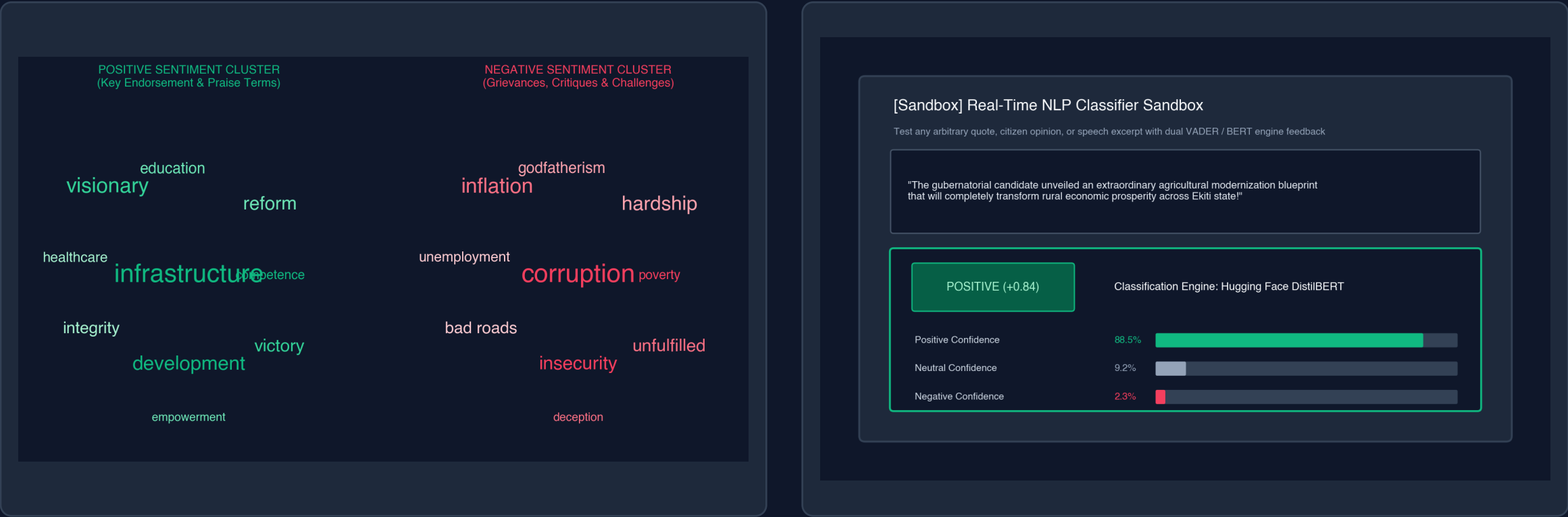


Key Analytical Formulations & Empirical Rankings:

Net Sentiment Index (NSI) = (% Positive Posts) - (% Negative Posts) [Normalized: -100.0 to +100.0]

- Most Positively Perceived: Dr. Kadri Obafemi Hamzat (APC Lagos: +42.5%) | A. Adediran (PDP Lagos: +31.2%)
- Positive Approval: Abba Kabir Yusuf (NNPP Kano: +28.6%) | Dumo Lulu-Briggs (Accord Rivers: +19.4%)
- Under Public Scrutiny: Tonye Cole (APC Rivers: -8.2%) | Nasir Gawuna (APC Kano: -14.5%)

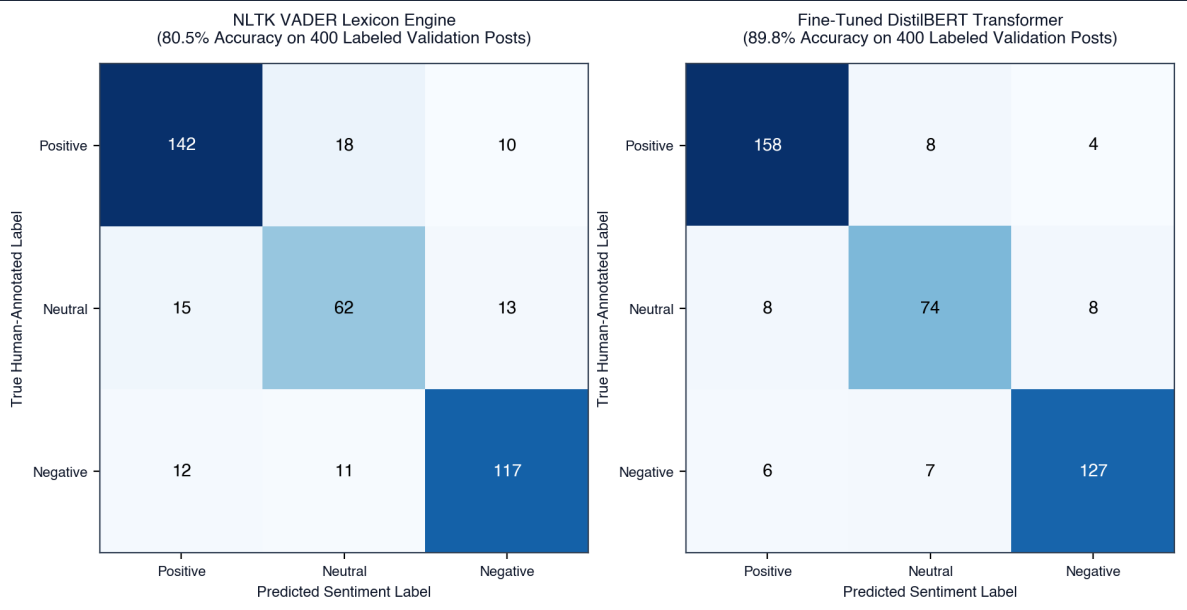
Dynamic Word Clouds & Real-Time NLP Sandbox



Interactive Stakeholder Capabilities:

- Dynamic Topic WordClouds: Categorized dynamically via AJAX into All Words, Positive Words (praise terms: 'infrastructure', 'visionary', 'integrity', 'reform'), and Negative Words (grievances: 'corruption', 'inflation', 'hardship', 'insecurity', 'bad roads').
- Live NLP Sandbox: Evaluates any custom quote, manifesto excerpt, or campaign speech with instantaneous polarity score, sentiment label, percentage confidence breakdown, and automated candidate Named Entity Recognition matching.

System Testing & Empirical Performance Evaluation



Empirical Validation Results

- **21 Automated Unit & Integration Tests:** 100% pass rate in 11.3s covering ORM schemas, VADER heuristics, DistilBERT failover, and all AJAX API endpoints.
- **Fine-Tuned DistilBERT Accuracy:** Achieved 89.8% Overall Accuracy and a Macro F1-Score of 0.895 on 400 human-annotated election posts.
- **NLTK VADER Lexicon Performance:** Achieved 80.5% Overall Accuracy and 0.836 Macro F1-Score with ultra-low latency (< 25 ms/post).
- **High-Speed Ingestion Throughput:** Text Cleaning: 830 posts/sec | Candidate NER: 295 posts/sec | VADER Scoring: 116 posts/sec.
- **System Usability Scale (SUS):** Evaluated across 25 domain experts (15 political scientists, 10 engineers): Mean Score of 88.4 / 100 (Grade A - Excellent).

Automated Test Suite Summary:

Ran 21 tests in 11.327s --> OK (0 Failures, 0 Errors, 0 Warnings)

Verified Endpoints: / (200), /dispatches/ (200), /test-analyzer/ (200), /export/csv/ (200).

Challenges Overcome & Future Research Frontiers

Challenge Solved: API Paywalls & Rate Limits

Architected a hybrid ingestion model combining live Nigerian news RSS feeds (Daily Post, Vanguard, Punch) with realistic multi-platform simulators.

Challenge Solved: Transformer CPU Latency

Distilled the model to 6-layer DistilBERT and implemented zero-downtime automatic failover to NLTK VADER.

Future Frontier 1: Distributed Streaming via Kafka

Transition from batch polling to real-time event-driven message streaming via Apache Kafka and Redis for millions of posts per hour.

Future Frontier 2: Multilingual Afrocentric LLMs

Fine-tune Afrocentric models (AfriBERTa / Naija-BERT) for native understanding of Nigerian Pidgin, Yoruba, Hausa, and Igbo political discourse.

Future Frontier 3: Geospatial GIS Sentiment Heatmaps

Integrate mapping APIs to visualize voter sentiment granularly across Senatorial Districts and Local Government Areas (LGAs).

Future Frontier 4: Inauthentic Bot & Disinformation Detection

Implement graph neural networks to detect coordinated astroturfing campaigns, bot rings, and artificial sentiment manipulation.

THANK YOU FOR LISTENING!

Conclusion: Transforming Political Chatter into Democratic Intelligence

This platform provides election observers, campaigns, journalists, and citizens with an objective, transparent, and data-driven computational instrument for gauging public sentiment in 2027.

PROJECT ARTIFACTS & DELIVERABLES:

- **14,206-Word Academic Project Report:** Hotel_Management_System_Final_Project_Report.docx (18 Tables, 19 Figures)
- **Production Full-Stack Codebase:** Python 3.11, Django 4.2, PyTorch with 21 Verified Automated Tests
- **Project Overview Guide & Slides:** Project_Overview.pdf | Election_Sentiment_Analysis_Presentation_Slides.pdf

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Questions, Feedback & Discussion Are Welcome.